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Respiratory viral detection in adult severe acute respiratory infection post-COVID-19 pandemic: implications for antimicrobial stewardship

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SUMMARY

Objectives: To characterise epidemiology, management, and outcomes of adult severe acute respiratory infection (SARI) in the post-COVID-19 era using virological surveillance and linked clinical data, and to explore implications for antimicrobial stewardship.
Methods: Prospective observational cohort study of adults hospitalised with SARI at a tertiary hospital in Glasgow, UK between January 2023-March 2024. Upper respiratory tract swabs underwent multiplex respiratory viral RT-PCR in addition to routine SARS-CoV-2 and influenza point-of-care testing (POCT), with linkage to routinely collected NHS data.
Results: Among 780 participants, respiratory samples were available for 676. ≥1 respiratory virus was detected in 46.5% by multiplex PCR, compared with 20.0% by routine POCT. SARS-CoV-2 and rhinovirus were most frequently detected. Chronic lung disease was associated with increased detection of viruses not included in routine testing. Empirical antibiotics was common (73% at admission). Positive SARS-CoV-2/influenza POCT at admission was associated with reduced odds of antibiotic prescribing (aOR 0.15, 95% CI 0.08–0.28) and shorter hospital stay.
Conclusion: Respiratory viruses were identified in nearly half of adult SARI, exceeding the yield of routine testing. Early virological diagnosis using POCT was associated with lower antibiotic use and shorter hospital stay. Expanded rapid respiratory viral testing may support antimicrobial stewardship but require further evaluation.
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Background

Before the COVID-19 pandemic, surveillance for adult severe acute respiratory infection (SARI) focused almost exclusively on influenza, even though influenza accounts for only 10–20% of influenza-like illness (ILI) and SARI presentations,[1–3] with the causative agent remaining unidentified in a high proportion of cases.

The emergence of SARS-CoV-2 in 2019, followed by the widespread implementation of non-pharmaceutical interventions (NPI), and increased influenza vaccination uptake, substantially reduced the circulation of endemic respiratory viruses.[4] As SARS-CoV-2 transitions to endemicity and NPIs have been withdrawn, future patterns of respiratory virus circulation and resulting aetiology of SARI remain uncertain.[5,6] The World Health Organisation (WHO) has recommended broadening SARI surveillance beyond influenza and SARS-CoV-2 to improve understanding of aetiology and support early detection of emerging respiratory threats.[7]

We previously implemented enhanced hospital SARI surveillance in Glasgow, UK.[8] Here, we report findings from a refined surveillance programme incorporating comprehensive virological testing and linked clinical data to characterise the epidemiology, management, and outcomes of adult SARI in the post-pandemic period.

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Methods

Study design and setting

We undertook a prospective observational cohort study of adults admitted with SARI to the Queen Elizabeth University Hospital (QEUH), a large tertiary referral centre in Glasgow, UK, that serves a population of 1.2 million with ~100,000 visits to the Emergency Department (ED) each year. The primary outcome was the weekly proportion of hospitalised SARI PCR-positive for SARS-CoV-2 or influenza A and B. Ethics approval was obtained from the North of Scotland Research Ethics Committee (Ref 21:/NS/1036) and the NHSGGC Biorepository (ref. 16/WS/0207) for storage of residual samples.

Recruitment and sampling

Eligible patients presenting to the ED or Medical Assessment Units (MAUs) were screened using a refined SARI definition aligned with Public Health Scotland criteria[9] (Table 1). Following the initial year of comprehensive recruitment,[8] enrolment was modified to incorporate three groups: Group 1: age-stratified sampling of SARI admissions (based on the age distribution of SARI patients recruited in Year 1)[8]; Group 2: patients with point-of-care (POC) polymerase chain reaction (PCR)-confirmed influenza A/B or SARS-CoV-2, and Group 3: patients requiring high dependency unit (HDU) or intensive care unit (ICU) support. This approach was designed to maintain recruitment of a representative sample of SARI admissions (Group 1) while enriching the cohort with patients with severe disease and clinically important outcomes in the context of a busy ED/MAU environment.

Routine virology testing of adult SARI cases in QEUH ED/MAUs comprised near-patient POC Reverse Transcription-PCR (RT-PCR) testing of combined nose and throat swabs for SARS-CoV-2 and influenza A/B using the Cobas® Liat® platform (Roche Diagnostics USA), with a turnaround time (TAT) of 20 min. With informed consent from patients, or a proxy, upper respiratory tract (URT) samples were collected from Groups 1–2. For Group 3, residual diagnostic samples were used when available. Samples were stored at −80°C in inactivation transport medium (ITM, Cambridge Bioscience) for multiplex PCR testing at a later date.

Data collection and linkage

At recruitment, we collected demographic data, vaccination status, presenting symptoms, vital signs, SARS-CoV-2/influenza point-of-care-test (POCT) result, and initial clinical management (Fig. 1). Following discharge, final diagnosis, antimicrobial and antiviral prescriptions, maximum level of care, respiratory support requirements, and clinical outcome were extracted from the electronic health record.

Data were entered into electronic case report forms (eCRF) (Castor, Netherlands), pseudonymised, and transferred to the NHS Greater Glasgow and Clyde (NHSGGC) SafeHaven Trusted Research Environment for linkage with routinely collected NHS datasets, comorbidities, Scottish Index of Multiple Deprivation (SIMD), medication history, laboratory, radiology, and mortality data (Fig. 1).

Laboratory methods

A multiplex RT-PCR panel, developed at the West of Scotland Specialist Virology Centre,[10] was used to test for adenovirus (HAdV), rhinovirus (HRV), respiratory syncytial virus (RSV) A/B, influenza A and B viruses (IAV and IBV), parainfluenza viruses (HPIV) 1–4, human coronaviruses (HCoV) NL63, 229E, and OC43, human metapneumovirus (HMPV), and *Mycoplasma pneumoniae (M.pneumoniae),* using a validated protocol.[11]

Statistical analysis

Categorical variables were summarised using frequencies and percentages; continuous variables as mean/standard deviation (SD) or median/interquartile range (IQR) as appropriate. Demographics, comorbidities, vaccination status, microbiology results, clinical presentation, antimicrobial use, length of hospital stay, clinical outcomes, and discharge diagnosis were compared by respiratory virus positivity. Further comparisons by critical care requirement and POCT positivity were carried out for some variables, as well as subgroup analyses by PCR-positivity for routinely tested-for viruses (i.e SARS-CoV-2/IAV/IBV) vs. other respiratory viruses (ORVs), individual virus positivity, virus mono-infections vs. viral coinfections, and viral vs. bacterial positivity. Differences were investigated by chi-square or Fisher exact test for categorical variables, and Kruskal-Wallis test

Table 1
SARI case definition, case inclusion and exclusion criteria, and recruitment groups.

SARI case definition	
Presenting with at least one of the following symptoms: • History of fever or measured fever of ≥38 °C • Cough • Shortness of breath Plus: • Onset of symptoms within the last 14 days AND • Presentation thought to be due to respiratory infection by the treating clinician AND • Requiring hospital admission	
Case Inclusion and Exclusion Criteria	
Case inclusion criteria	• Aged 16 years or over • Meets the SARI case definition • Admitted to the hospital within 24 h of potential enrolment
Case exclusion criteria	• Presentation with an acute respiratory illness not felt to be due to infection • Discharged from ED/MAU • Tested for influenza ± SARS-CoV-2 > 24 h after hospital admission • Recent discharge from hospital (within 14 days)
Recruitment Groups	
Group 1	Age-stratified daily sampling; 2 SARI patients daily/10 per week in the following age groups*: a) 16–64 years, b) 65–79 years, and c) > 80 years.
Group 2	SARI patients PCR-positive for SARS-CoV-2 or influenza A/B on point-of-care testing in ED/MAU and not recruited to Group 1.
Group 3	SARI patients admitted to critical care who were not recruited to Groups 1 or 2.

Abbreviations: SARI, severe acute respiratory infection; ED, emergency department; MAU, medical admissions unit; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; QEUH, Queen Elizabeth University Hospital.

* Based on the first year of recruitment,[8] recruitment of similar numbers of the defined age groups would provide a representative sample of SARI cases presenting to QEUH.

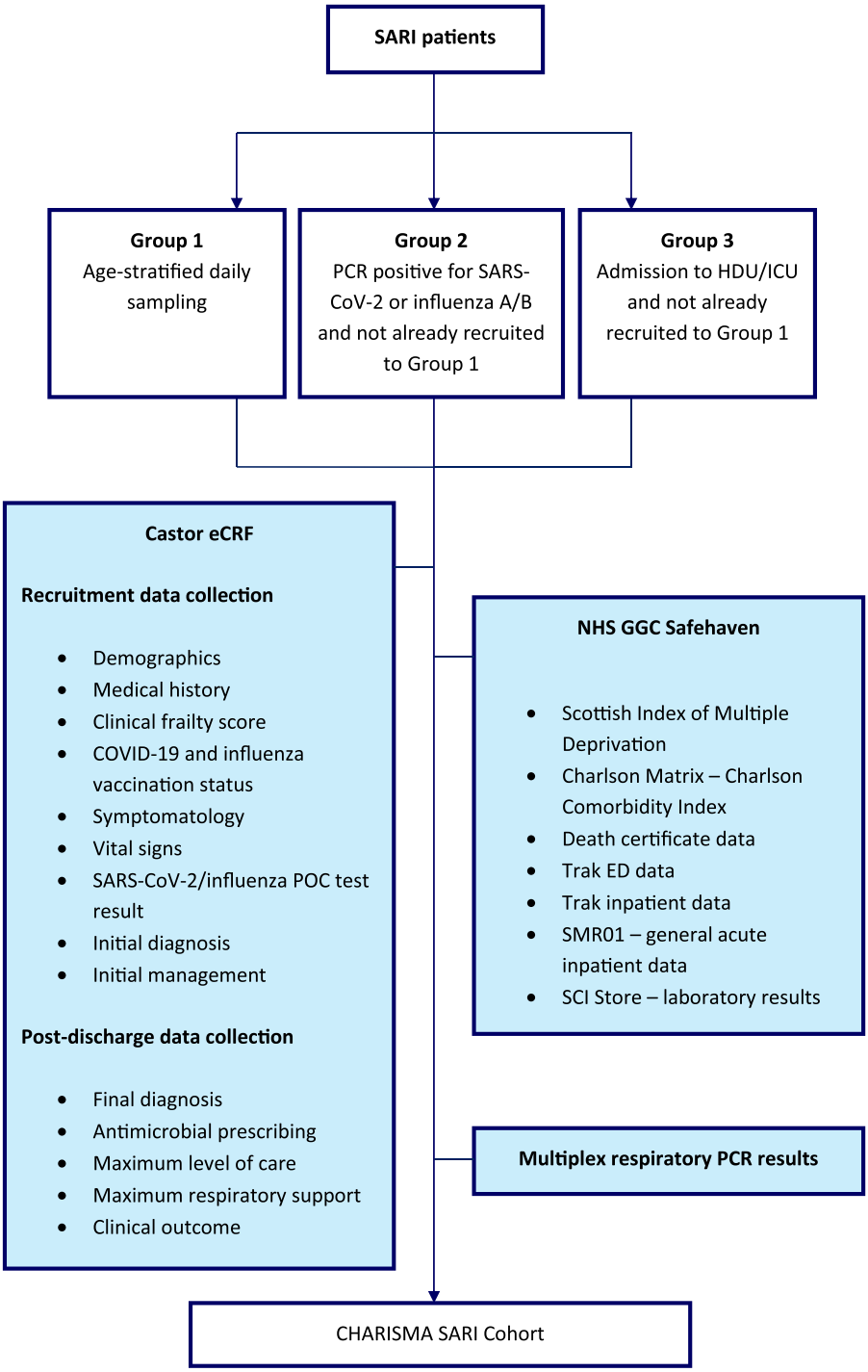


Fig. 1. Data linkage diagram.
Abbreviations: SARI, severe acute respiratory infection; PCR, polymerase chain reaction; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; HDU, high dependency unit; ICU, intensive care unit; eCRF, electronic case report form; COVID-19, coronavirus disease 2019; POC, point of care; ED, emergency department.

for continuous variables. P≤0.05 was considered statistically significant. Binomial confidence intervals were calculated for proportions. Some analyses were restricted to certain groups to avoid bias introduced by group recruitment criteria, e.g. factors associated with POC and multiplex PCR positivity (Groups 1 and 3), factors associated with clinical outcomes (Group 1), and clinical outcomes by virus positivity (Groups 1 and 2).

Receipt of one or more doses of COVID-19 vaccine at any time was classified as 'vaccinated'. One dose of the current season's influenza vaccine was classified as 'vaccinated'. Consolidation (or equivalent descriptor) on chest imaging (chest X-ray or computerised tomography (CT)) report conducted between −1 to +3 days of admission was classified as 'consolidation'.

An appropriate microbiology sample for SARI was defined as a blood culture, lower respiratory tract (LRT) sample, or urinary legionella antigen test (LAT), collected between −7 to +7 days of admission. Coagulase-negative *staphylococci* in blood/LRT sample culture, or any *Candida* species from LRT samples, were deemed insignificant and excluded. All other identified bacteria were deemed significant. For selected comparative analyses of viral and bacterial

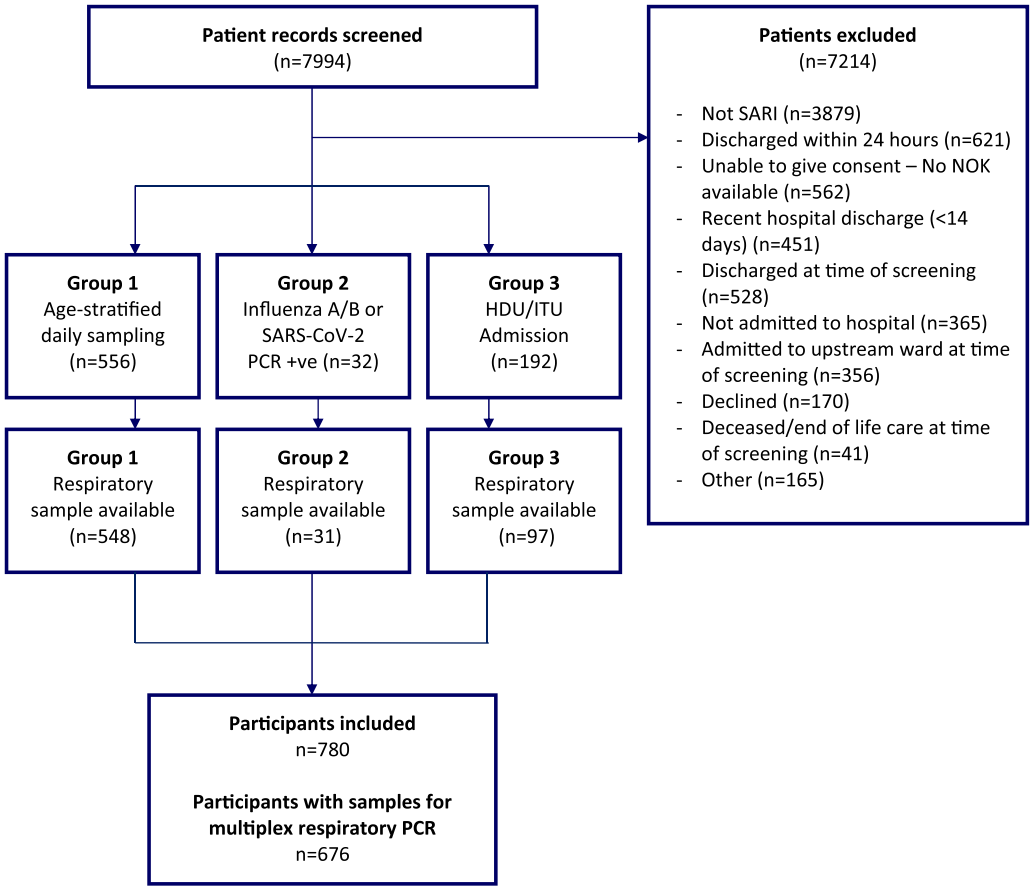


Fig. 2. Patients screened and recruited.
Abbreviations: SARI, severe acute respiratory infection; NOK, next of kin; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; PCR, polymerase chain reaction; HDU, high dependency unit; ICU, intensive care unit.

infections, participants with confirmed viral–bacterial co-infection were excluded. Haematology and biochemistry results were categorised as high, normal, or low based on cut-off values used in the NHSGGC laboratories (Supplementary Table 1).

Mixed-effects logistic regression identified predictors of positivity for SARS-CoV-2/influenza infection versus other respiratory viruses (ORVs), with the participant as a random effect, and a forward stepwise selection process to identify factors for inclusion. The variables available for selection included patient demographics, comorbidities, and vaccination status. Multivariable logistic regression examined predictors of in-hospital mortality and antibiotic prescribing. For participants contributing multiple SARI episodes, only the first episode was included. Age-group was selected a priori; other variables available for backward stepwise selection were as above plus virological factors, admission clinical parameters, and investigation results. The effect of POCT positivity on length of hospital admission was modelled using an accelerated failure time (AFT) framework, adjusting for age and number of comorbidities. All analyses were performed with R version 4.3.1 using *tidyverse*, *MASS, lme4,* survival, and *finalfit.*

Results

Participant characteristics

From 16th January 2023 to 28th March 2024 (with a planned pause July to September 2023), 780 adult SARI admissions were included (Fig. 2), comprising 556 from Group 1 (71.3%), 32 from Group 2 (4.1%), and 192 from Group 3 (24.6%). Respiratory samples were available for 676 participants, comprising 548 (81.1%) from Group 1, 31 (4.6%) from Group 2, and 97 (14.3%) from Group 3. Twenty-one participants contributed to multiple SARI episodes during the study period.

Median age was 64 years (IQR 51–74 years), 58.5% were female (n=456), 93.3% (n=728) were white, and nearly two-thirds were from the two most deprived SIMD quintiles (63.0%, n=482; Table 2). Critical care patients were younger (median age 60 years (IQR 45.0–70.8) vs 66 years (54.2–75.0, $p < 0.001$), more often male (52.9% vs 37.4%, $p < 0.001$), more likely of non-white ethnicity (p=0.02), and more frequently from a deprived quintile (p=0.03).

Vaccination coverage was high overall (COVID-19 ≥1 dose: 93.4% (n=720), ≥3 doses 82.9% (n=639); current season influenza vaccine: 50.6% (n=391)) (Table 2), but lower in critical care patients for both COVID-19 (90.2% vs 94.5%, p=0.049) and influenza (42.2% vs 53.7%,p=0.006).

Comorbidity burden was substantial; 77.5% (n=527) had at least one long-term condition, with a median Charlson Comorbidity Index (CCI) score of 4 (IQR 2–6). Chronic lung disease (see Supplementary Table 2 for included ICD-10 codes) was the most common (54.1%, n=400), followed by malignancy (16.4%, n=121), diabetes (11.9%, n=88), and heart failure (11.2%, n=83) (Table 2).

Viral aetiology

Among participants in Groups 1 and 3, POC PCR positivity was 13.0% for SARS-CoV-2 (97/748, 95% CI 10.6–15.6%), 6.0% for IAV (45/748, 95% CI 4.4–8.0%) and 1.5% for IBV (11/748, 95% CI 0.7–2.6%). Temporal trends aligned with national patterns of COVID-19 and

Table 2
Baseline characteristics of participants.

	N (%)	All	Ward (n=570)	Critical Care (n=210)	p-value**
Demographics					
Age (years)	780 (100.0)				
Median (IQR)		64.0 (51.0 to 74.0)	66.0 (54.2 to 75.0)	60.0 (45.0 to 70.8)	< 0.001
Age Group (years)	780 (100.0)				
16–49		174 (22.3)	111 (19.5)	63 (30.0)	< 0.001
50–64		218 (27.9)	148 (26.0)	70 (33.3)	
65–79		293 (37.6)	232 (40.7)	61 (29.0)	
≥80		95 (12.2)	79 (13.9)	16 (7.6)	
Sex	780 (100.0)				
Female		456 (58.5)	357 (62.6)	99 (47.1)	< 0.001
Ethnicity	780 (100.0)				
African, other, or mixed ethnic groups		13 (1.7)	6 (1.1)	7 (3.3)	0.02
Asian or Asian Scottish/British		30 (3.8)	20 (3.5)	10 (4.8)	
White		728 (93.3)	540 (94.7)	188 (89.5)	
Scottish Index of Multiple Deprivation 2020 Quintile	680 (87.2)				
1 (most deprived)		276 (40.6)	192 (38.4)	84 (46.7)	0.031
2		152 (22.4)	110 (22.0)	42 (23.3)	
3		59 (8.7)	42 (8.4)	17 (9.4)	
4		78 (11.5)	58 (11.6)	20 (11.1)	
5 (least deprived)		115 (16.9)	98 (19.6)	17 (9.4)	
Recruitment cohort	780 (100.0)				
Group 1: Age-stratified daily sampling		556 (71.3)	*	*	< 0.001
Group 2: PCR+ for Flu A/B or SARS-CoV-2		32 (4.1)	*	*	
Group 3: Admitted to HDU or ICU		192 (24.6)		192 (91.4)	
Vaccination Status					
Covid-19 Vaccination	771 (98.8)				
Vaccinated		720 (93.4)	536 (94.5)	184 (90.2)	0.049
Influenza Vaccination	772 (99.0)				
Vaccinated		391 (50.6)	304 (53.7)	87 (42.2)	0.006
Comorbidities					
Number of comorbidities	680 (87.2)				
0		153 (22.5)	109 (21.8)	44 (24.4)	0.765
1		265 (39.0)	197 (39.4)	68 (37.8)	
≥2		262 (38.5)	194 (38.8)	68 (37.8)	
Charlson Comorbidity Index	690 (88.5)				
0–2		212 (30.7)	144 (28.5)	68 (37.0)	0.042
3–5		216 (31.3)	170 (33.6)	46 (25.0)	
≥5		262 (38.0)	192 (37.9)	70 (38.0)	
HIV infection	739 (94.7)	*	*	*	0.071
Cerebrovascular Disease	739 (94.7)	68 (9.2)	49 (9.0)	19 (9.6)	0.915
Heart Failure	739 (94.7)	83 (11.2)	58 (10.7)	25 (12.7)	0.532
Diabetes	739 (94.7)	88 (11.9)	60 (11.1)	28 (14.2)	0.299
Cancer	739 (94.7)	121 (16.4)	84 (15.5)	37 (18.8)	0.34
Myocardial Infarction	739 (94.7)	78 (10.6)	64 (11.8)	14 (7.1)	0.088
Liver Disease	739 (94.7)	58 (7.8)	47 (8.7)	11 (5.6)	0.22
Lung Disease	739 (94.7)	400 (54.1)	311 (57.4)	89 (45.2)	0.004
Peripheral Vascular Disease	739 (94.7)	42 (5.7)	28 (5.2)	14 (7.1)	0.408
Chronic Kidney Disease	739 (94.7)	66 (8.9)	47 (8.7)	19 (9.6)	0.792
Rheumatological Disease	739 (94.7)	47 (6.4)	39 (7.2)	8 (4.1)	0.17
Dementia	780 (100.0)	5 (0.6)	*	*	1.00
Peptic Ulcer Disease	780 (100.0)	28 (3.6)	20 (3.5)	8 (3.8)	1.00
Clinical Frailty Category	776 (99.5)				
1–3. Not frail		423 (54.5)	327 (57.4)	96 (46.6)	0.009
4–6. Mild-moderate frailty		306 (39.4)	215 (37.7)	91 (44.2)	
7–9. Severe frailty		47 (6.1)	28 (4.9)	19 (9.2)	

Abbreviations: IQR, interquartile range; GP, general practitioner; PCR, polymerase chain reaction; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; Flu, influenza virus; HDU, high dependency unit; ICU, intensive care unit; COVID-19, coronavirus disease 2019; HIV, human immunodeficiency virus.

* Indicates values that have been suppressed due to the potential risk of disclosure and to help maintain confidentiality

** Chi-squared or Fisher's exact test depending on expected frequencies > 5 or < 5 respectively if categorical data. Kruskall-Wallis test if continuous data.

influenza hospital admissions[12] (Fig. 3a & b). Most IAV detections (80%) occurred between December 2023 and February 2024 (Fig. 3b). IBV was identified sporadically (February–October 2023) but was absent during the IAV peak.

Across Groups 1 and 3, multiplex PCR identified at least one respiratory virus in 46.5% cases (300/645; 95% CI, 42.6–50.4%). SARS-CoV-2 was the most frequently detected (13.3%, n=86), followed by HRV (11.0%, n=71), IAV (5.9%, n=38), HMPV (4.3%, n=28), RSV (3.9%, n=25), and HPIV3 (3.3%, n=19) (Fig. 4a). HRV circulated year-round with a peak in November 2023. HPIV3 was predominantly observed in March-June 2023, RSV in October 2023-January 2024, and *M.pneumoniae* from October 2023-March 2024. Multiplex testing identified respiratory viruses in an additional 171 SARI patients compared with SARS-CoV-2/Influenza POCT (46.5% vs. 20.0%, p < 0.001).

Age distributions differed by virus; for example, SARS-CoV-2, and HPIV3 mono-infections were more common in older adults (median age 72.5 and 68.0 years, respectively), while IBV and *M. pnuemoniae* were mainly detected in younger adults (median ages 34 years for both). COVID-19 and influenza vaccination rates did not differ by SARS-CoV-2 or IAV status. However, IBV-positive patients were less frequently influenza-vaccinated (16.7% vs 51.2%, p=0.02).

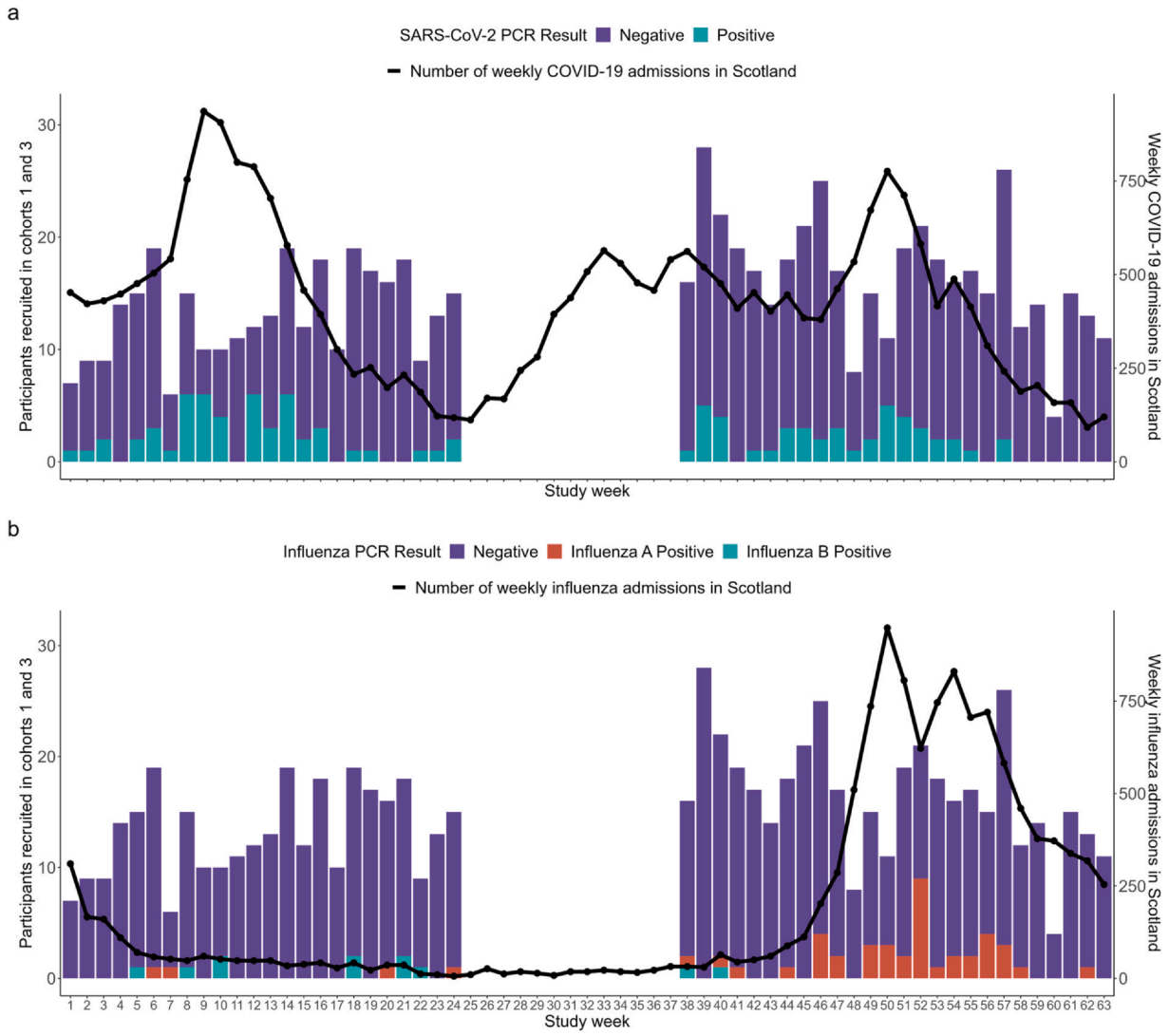


Fig. 3. Number of SARI patients recruited weekly (Groups 1 and 3) with proportion positive on a) SARS-CoV-2 point-of-care PCR and weekly Scottish COVID-19 admissions data; b) Influenza A/B point-of-care PCR and weekly Scottish influenza admissions data. Study week number represents the week of the study starting from 16th January 2023 in week 1 and ending on 28th March 2024 in week 63. Weeks 25-37 represent the planned break in recruitment in summer 2023. Weekly admissions data taken from Public Health Scotland (11)

In multivariable mixed effects analysis of PCR-positive cases, chronic lung disease was associated with an increased odds of infection with ORVs compared to SARS-CoV-2/IAV/IBV mono-infections (adjusted odds ratio (aOR) 1.64, 95%CI 1.00–2.71, p=0.05). Adults aged ≥80 years had a lower odds of ORV infection (aOR 0.35, 95% CI 0.15–0.82, p=0.02) (Supplementary Table 3).

Viral co-infection

Among Group 1 and 3 participants, respiratory virus co-infections were detected in 3.7% (24/645; 95% CI 2.4–5.5%), with higher rates in critical care than ward-level patients (6.4% (7/109) vs 3.2% (17/536). Co-infections comprised 11.6% of SARS-CoV-2 infections and 18.4% of IAV infections (Supplementary Figure 1). Compared to mono-infections, co-infected patients were younger (52.5 vs. 63.9 years, p=0.01), less likely to have chronic lung disease (20.0% vs. 56.1%, p=0.001), and less likely to be influenza vaccinated (19.2% vs. 54.1%, p=0.001) (Supplementary Table 4).

Bacterial aetiology

At least one appropriate microbiology sample was obtained in 58.7% of SARI episodes (458/780), including LRT specimens (31.8%, n=248), blood cultures (40.1%, n=316), and urinary LAT (23.1%, n=180). Sampling was more frequent in critical care patients than ward patients (90.5% vs. 47.0%). Most LATs (84.4%, 152/180) were performed in critical care, and none were positive.

Of the 458 patients with appropriate microbiology sampling, 21.4% (n=98) had ≥1 significant organism identified, with higher positivity in critical care patients (29.5% vs.15.7%, p=<0.001). The most frequently pathogens were *Haemophilus influenzae* (7.1%, n=31), *Streptococcus pneumoniae* (2.5%, n=11), and *Staphylococcus aureus* (2.3%, n=10) (Fig. 4b). *H. influenzae* and *S. aureus were* mainly identified in LRT samples (96.8% and 90%, respectively), while *S. pneumoniae* was primarily identified in blood cultures (72.7%).

Among those tested with both multiplex PCR and microbiology sampling (n=368), bacterial positivity did not differ between virus-positive and virus-negative cases (17.1% vs. 20.7%, p=0.46). Microbiology samples were positive in 20.9% (95%CI 10.0–36.0%) of

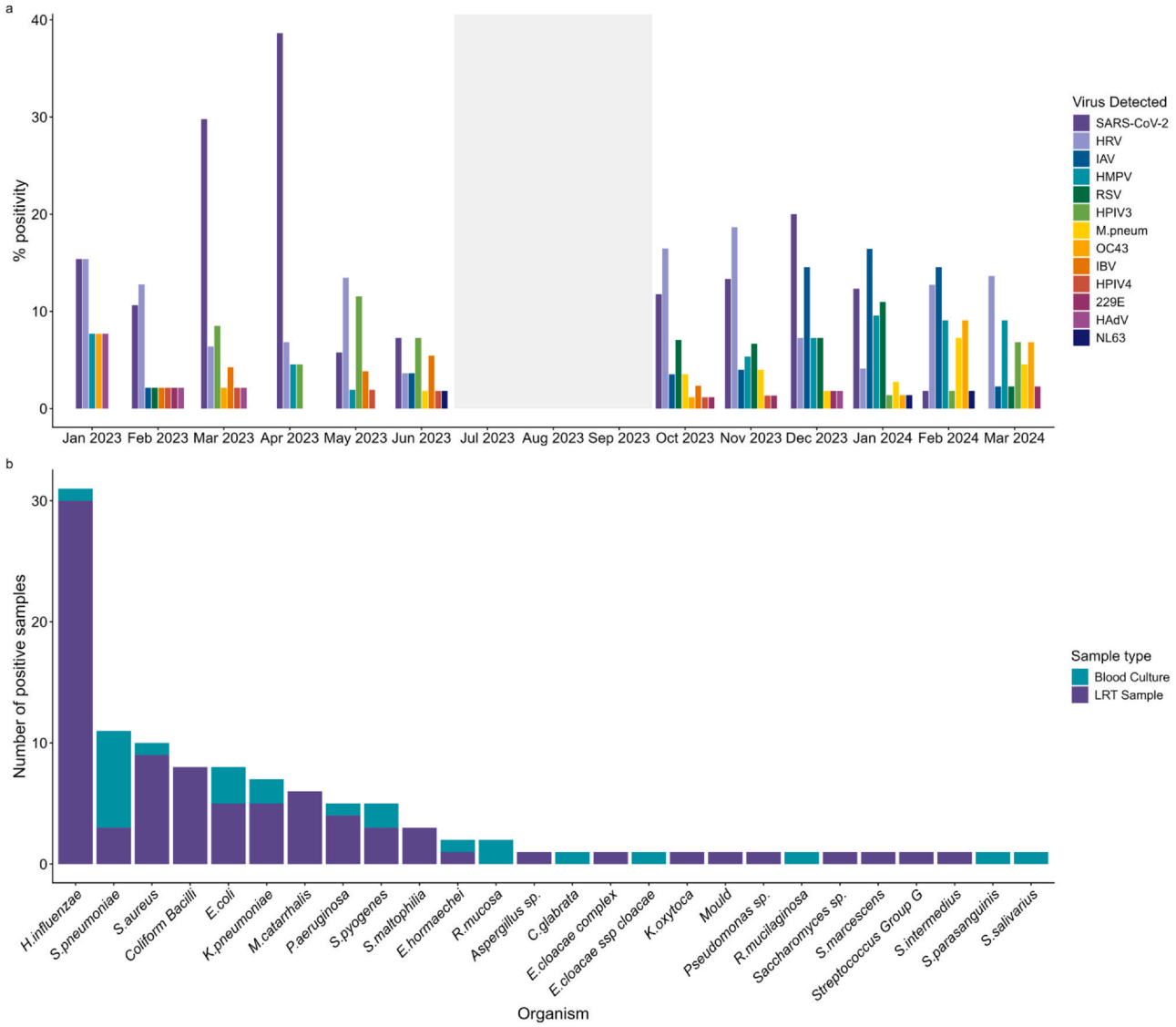


Fig. 4. Viral and bacterial aetiology of SARI cases, a) Respiratory viruses identified by multiplex respiratory PCR of upper respiratory tract swabs from Groups 1 and 3 by month; b) Significant bacterial pathogens identified from blood cultures and lower respiratory tract samples in recruited SARI patients.
Abbreviations: SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; HRV, rhinovirus; IAV, influenza A virus ; HMPV, human metapneumovirus; RSV, respiratory syncytial virus; HPIV3, parainfluenza 3 virus; M.pneum, Mycoplasma pneumoniae, OC43, human coronavirus OC43; IBV, influenza B virus; HPIV4, parainfluenza 4 virus; HAdV, adenovirus; NL63, human coronavirus NL63; 229E, human coronavirus 229E.

SARS-CoV-2 mono-infected participants and 16.6% (95%CI 3.5–41.4%) of IAV mono-infected participants.

Clinical presentation

Median symptom duration before admission was 5 days (IQR 3–7). Common presenting symptoms were shortness of breath (90.8%, n=708), cough (85.1%, n=664), fatigue (71.2%, n=555), and fever (58.2%, n=454). Overall, 38.0% (292/769) required oxygen therapy, and 46.2% (348/753) had radiological consolidation.

Laboratory abnormalities at admission included raised white cell count (62.4%, 482/773), neutrophilia (68.4%, 529/773) lymphopenia (37.1%, 287/773), elevated C-reactive protein (CRP) (85.6%, 664/776), elevated lactate (39.6%, 194/490) and deranged renal function (32.7%, 254/776).

Compared to virus-negative patients, multiplex viral PCR-positive cases more frequently reported cough (92.7% vs. 82.6%, p < 0.001), fever (65% vs. 55.7%, p=0.008), sputum production (61.9% vs. 52.2%, p=0.001), rhinorrhoea (57.1% vs. 40.3%, p= < 0.001), and myalgia (52.4% vs. 42.6%, p=0.02), and were less likely to have consolidation (36.9% vs. 47.9%, p=0.006) (Supplementary Figure 2a). Lymphopenia was more common in virus-positive patients (41.3% vs. 33.1%, p=0.04) (Supplementary Figure 2b).

Excluding confirmed viral-bacterial co-infection, bacterial infections was associated with higher rates of neutrophilia (81.1% vs. 62.6%, p=0.03), consolidation (56.9% vs. 38.0%, p=0.04), and abnormal clotting (30.2% vs. 14.6%, p=0.03), while viral infections were more strongly associated with upper respiratory tract symptoms, e.g. cough (91.1% vs. 73.6%, p=0.005), sore throat (50.0% vs. 32.1%, p=0.04), and rhinorrhoea (57.3% vs. 34.0%, p=0.007). (Supplementary Table 5).

Antimicrobial and antiviral prescriptions

Pre-hospital antibiotics were administered to 25.3% (194/766) of SARI cases. At admission, empirical antibiotics were prescribed in 73.0% (568/778) of participants, antivirals in 4.2% (33/777), and systemic steroids in 40.2% (312/777; 43.4% of SARS-CoV-2 PCR-

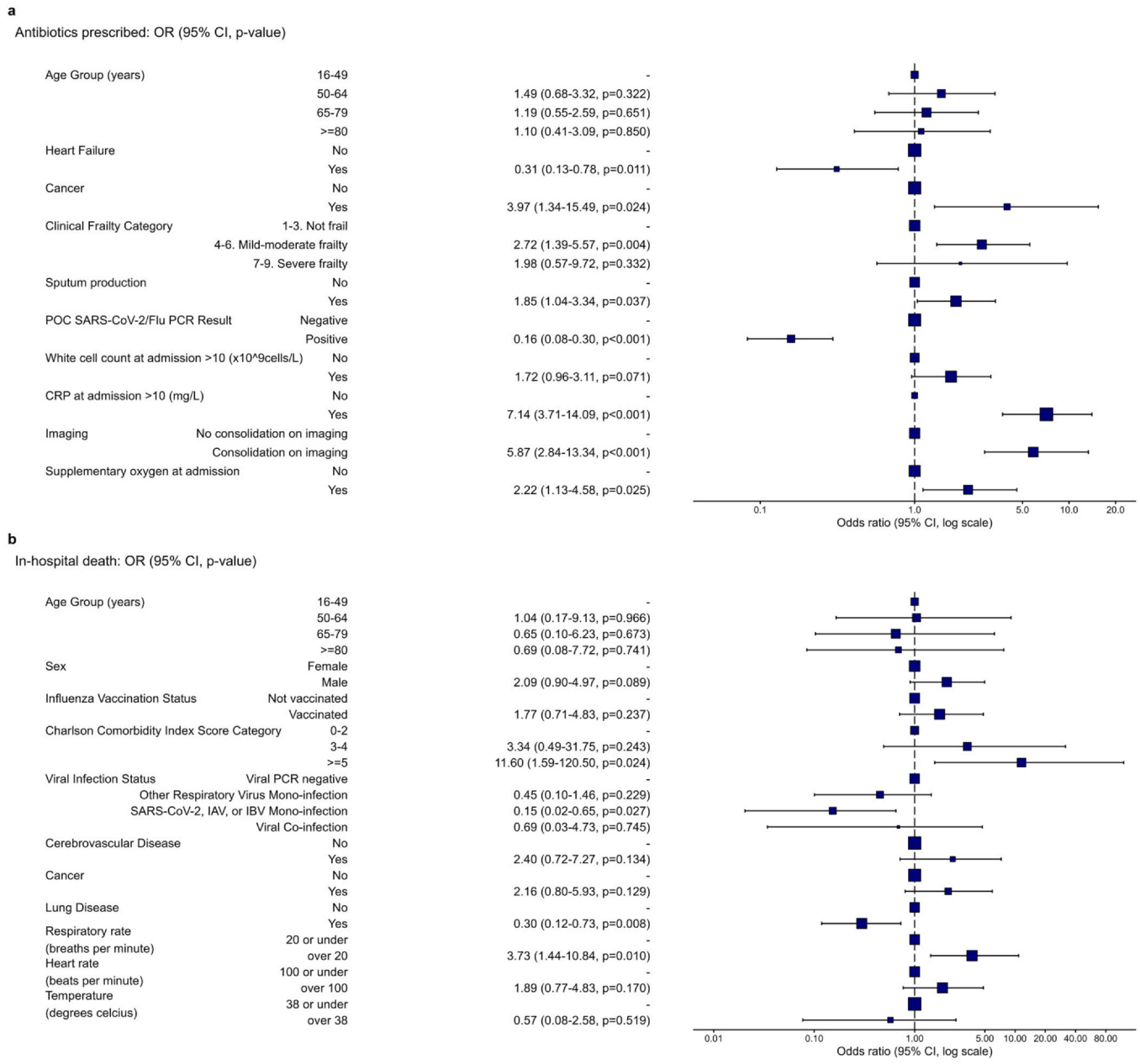


Fig. 5. Factors associated with a) antibiotic prescribing and b) in-hospital death.

positive participants, 63.6% of those diagnosed with an exacerbation of COPD). During the inpatient stay, this rose to 85.5% (664/777) for antibiotics, 11.6% (90/777) for antivirals, and 58.6% (455/777) for systemic steroids (63.9% of SARS-CoV-2 cases and 95.3% COPD exacerbations).

Only 62.6% (432/690) of those prescribed antibiotics had an appropriate microbiological sample taken. Amoxicillin (42.0%, 327/778), clarithromycin (26.6%, 207/778), doxycycline (13.6%, 106/778), and co-amoxiclav (11.8%, 92/778) were the most commonly used agents (Supplementary Figure 3a); 19.7% received amoxicillin-clarithromycin dual therapy.

A positive SARS-CoV-2 or influenza POCT was associated with significantly lower antibiotic prescription in ED (52.8% vs. 79.1%, $p < 0.001$) and as inpatients (68.2% vs. 90.6%, $p < 0.001$). Consolidation on imaging was associated with increased inpatient antibiotic use (95.4% vs. 77.5%, $p < 0.001$) and greater microbiology sampling (72.4% vs. 47.4%, $p < 0.001$), though culture positivity did not differ (23.0% vs. 18.2%, p=0.27).

Critical care patients were more likely to receive inpatient antibiotics (91.8% vs. 83.2%, p=0.004), including more broad-spectrum agents (co-amoxiclav: 51.7% vs.17.4%, piperacillin-tazobactam: 34.8% vs. 7.2%, both $p < 0.001$) (Supplementary Figure 3b).

Independent predictors of antibiotic prescription included a raised admission CRP (aOR 7.14, 95% CI 3.71–14.09 $p < 0.001$), radiological consolidation (aOR 5.87, 95% CI 2.84–13.34, $p < 0.001$), supplemental oxygen required at admission (aOR 2.22, 95% CI 1.13–4.58 p=0.03), and a history of malignancy (aOR 3.97, 95% CI 1.34–15.49, p=0.02) (Fig. 5a). Conversely, a history of heart failure (aOR 0.31, 95% CI 0.13–0.78, p=0.01) and a positive SARS-CoV-2/Influenza POCT (aOR 0.16, 95% CI 0.08–0.30, $p < 0.001$) were associated with lower odds of antibiotics prescription.

Among influenza PCR-positive participants (n=63), 77.8% received oseltamivir. Among SARS-CoV-2 PCR-positive participants (n=122), 63.9% received steroids, 18.2% received remdesivir (13/23 critical care-admitted) and 5.7% received immunomodulators (tocilizumab or baricitinib) (5/7 in critical care).

Table 3
Final diagnosis at discharge from hospital for all SARI cases including proportions with respiratory virus detected and antibiotics prescribed.

Diagnosis	Participants with diagnosis (n (%))	Virus-positive participants by diagnosis (n (%))	Antibiotic prescriptions by diagnosis (n (%))	Antibiotic prescriptions in virus positive participants by diagnosis (n (%))
LRTI /Community acquired pneumonia	361 (46.3%)	120 (33.2%)	355 (98.3%)	116 (96.7%)
COPD exacerbation	151 (19.4%)	54 (35.8%)	132 (87.4%)	46 (85.2%)
COVID-19	120 (15.4%)	116 (96.7%)	80 (66.7%)	77 (66.4%)
Asthma exacerbation	93 (11.9%)	58 (62.4%)	69 (74.2%)	46 (79.3%)
Other diagnosis	68 (8.7%)	27 (39.7%)	60 (88.2%)	23 (85.2%)
Influenza infection	64 (8.2%)	64 (100%)	38 (59.4%)	38 (59.4%)
Other respiratory condition	45 (5.8%)	8 (17.8%)	42 (93.3%)	7 (87.5%)
Other respiratory infection	30 (3.8%)	11 (36.7%)	25 (83.3)	9 (81.8%)
Other cardiac condition	30 (3.8%)	11 (36.7%)	23 (76.7%)	9 (81.8%)
Cardiac failure	22 (2.8%)	9 (40.9%)	20 (90.9%)	8 (88.9%)
Other infection	22 (2.8%)	7 (31.8%)	19 (86.4%)	6 (85.7%)
Pulmonary embolism	16 (2.1%)	*	12 (75.0%)	*
Aspiration pneumonia	7 (0.9%)	*	7 (100%)	*

Abbreviations: LRTI, lower respiratory tract infection; COPD, chronic obstructive pulmonary disease
* Indicates values that have been suppressed due to the potential risk of disclosure and to help maintain confidentiality

Clinical outcomes

Clinical outcomes were analysed in Group 1 (n=556). Overall, 2.9% (n=16) required critical care. Oxygen therapy was required in 60.2% (n=346) needed supplemental oxygen, but very few required advanced respiratory support. In-hospital mortality was 2.5% (n=14). Median length of stay among survivors was 5 days (IQR 3–8).

In Groups 1–2, maximum respiratory support, critical care admission, and mortality did not differ by virus status. However, virus-positive patients had shorter hospital stay than virus-negative patients (median 5 vs. 6 days, p=0.002). A positive admission SARS-CoV-2/influenza POC PCR was associated with an even shorter stay (4 vs. 5 days, p=0.003). In a log-logistic accelerated failure time model, POCT-positive patients had a 23% shorter length of stay (time ratio = 0.77, 95% CI 0.63–0.93, p=0.007), adjusting for age and number of comorbidities (Supplementary Table 6).

Across the full cohort, the most common discharge diagnoses were LRTI/community acquired pneumonia (CAP) (46.3%, n=361), COPD exacerbation (19.4%, n=151), COVID-19 (15.4%, n=120), and asthma exacerbation (11.9%, n=93) (Table 3). Viral infections were frequently identified within these groups, including 62.4% of asthma exacerbations, 35.8% of COPD exacerbations, and 33.2% of LRTI/CAP cases.

Antibiotic prescribing varied by diagnosis; lowest in those with a discharge diagnosis of influenza (59.4%) and COVID-19 (66.7%), but very high for COPD (87.4%,) and asthma (74.2%) exacerbations, despite frequent viral aetiology (Table 3).

In multivariable analysis, in-hospital death was associated with a CCI ≥5 (aOR 11.60, 95% CI 1.59–120.50 p=0.02), and tachypnoea at admission (aOR 3.73, 95% CI 1.44–10.84, p=0.01). SARS-CoV-2/influenza mono-infection (aOR 0.15, 95% CI 0.02–0.65, p=0.03) and a history of chronic lung disease (aOR 0.30, 95% CI 0.12–0.73, p=0.01) were associated with lower mortality (Fig. 5b).

Discussion

In this prospective post-pandemic adult SARI cohort, nearly half of hospitalised patients undergoing multiplex PCR had at least one respiratory virus identified, more than double the diagnostic yield of routine POCT. SARS-CoV-2 remained a major contributor to SARI admissions, co-circulating with influenza and year-round rhinovirus. RSV, HMPV and PIV3 demonstrated seasonal patterns similar to the pre-pandemic period, while a marked increase of *Mycoplasma pneumoniae* was observed in early 2024, consistent with national and international reports.[12,13] Importantly, a positive influenza or SARS-CoV-2 POCT result at admission was associated with reduced antibiotic prescribing and shorter hospital stay, highlighting the potential clinical relevance of early virological diagnosis in acute respiratory illness.

SARS-CoV-2 and rhinovirus were the most frequently detected viruses, consistent with recent studies from Norway[14] and Japan,[15] but contrasting to reports from China where influenza A predominated.[16–20] Such differences likely reflect variations in study periods, population demographics, healthcare pathways, and regional patterns of viral circulation. Seasonal dynamics also varied by virus; in the UK, influenza A showed a pronounced winter peak, whereas SARS-CoV-2 exhibited broader activity with peaks in both spring and winter during the study period. Whether SARS-CoV-2 will ultimately establish a stable seasonal pattern remains uncertain. Notably, influenza and SARS-CoV-2 co-circulated during winter 2023–2024, adding further pressure to healthcare services. RSV, rhinovirus, HMPV and HPIV3 followed expected pre-pandemic seasonal patterns,[21,22] and we captured the widely reported resurgence of *M.pneumoniae,* which began globally in summer 2023,[13] peaking locally and nationally in February 2024.[12] Nonetheless, detection of respiratory viruses alone should not be interpreted as evidence of causation, given the potential for incidental carriage, prolonged viral shedding, or co-infection.

Viral co-infections were uncommon and occurred predominantly in younger adults with fewer comorbidities. This proportion lies at the lower end of post-pandemic estimates, likely reflecting the older age profile of our SARI cohort, as viral co-infections are more frequently reported in children and young adults.[23] Co-infections most often involved influenza A virus or SARS-CoV-2. The clinical significance of specific viral combinations remains uncertain, with existing data showing mixed effects on disease severity,[23] and warrants further investigation.

Consistent with a broad range of recent international studies,[14–16,24–29] multiplex PCR detected at least one respiratory virus in approximately half of tested SARI cases, while routine testing alone failed to identify more than half of these infections. Despite limited microbiological evidence of bacterial infection, empirical antimicrobial use remained high, highlighting persistent challenges of antimicrobial stewardship in acute respiratory illness. Early diagnosis of influenza and SARS-CoV-2 via POCT was associated with both reduced antibiotic prescribing and shorter length of stay. Although prior CAP and SARI studies have reported heterogeneous effects of POCT on clinical management,[30–35] reductions in antibiotic use[31,32] and/or hospital stay[32] have been observed following positive POCT results. Studies reporting no effect[30,33–35] mostly pre-dated SARS-CoV-2 and/or used POCTs with a TAT of at least 45 min, which is likely too slow to impact on antibiotic prescribing. POCTs with a 15–20-minute TAT, are now available which warrant further evaluation.

Our findings suggest that expanded rapid respiratory viral testing at admission, either universally or targeted to groups with high diagnostic yield, may inform antimicrobial stewardship strategies. Patients with chronic lung disease represent a particularly relevant group, as they are frequently virus-positive, including for pathogens not currently included in routine testing, yet commonly receive antibiotics for COPD and asthma exacerbations despite predominantly viral aetiology. This is consistent with prior studies demonstrating substantial antibiotic exposure in these syndromes.[36–38] Moreover, detection of pathogens such as *M. pneumoniae*, which has specific therapeutic implications, further supports the potential added clinical value of broader panels.

Strengths of our study include its prospective design, comprehensive clinical and microbiological data, broad virological testing and secure linkage to routine healthcare datasets. Limitations included its single-centre nature, which may limit generalisability, although the study was conducted at Scotland's largest tertiary referral hospital that serves a socio-economically diverse population. Recruitment paused during summer months, limiting continuous year-round surveillance, and children were not included. As an observational study, associations between viral detection, antibiotic prescribing and length of stay may partly reflect clinician behaviour in response to test results rather than direct causal effects.

In conclusion, respiratory viruses were identified in nearly half of adults hospitalised with SARI in the post-pandemic period, substantially exceeding the diagnostic yield of current POCT strategies. Expanded viral POCT, particularly when targeted to patients with chronic lung disease, may support antimicrobial stewardship and patient flow, although confirmation in multi-centre studies with formal cost effectiveness analyses is required. Ongoing syndromic surveillance integrating clinical, microbiological and virological data remains essential as respiratory viral epidemiology continues to evolve and new preventative interventions are implemented, including RSV vaccination programmes for pregnant women, adults aged $\geq$ 75 years, adults aged 65–74 with chronic respiratory disease or immunosuppression, and care home residents in the UK.

CRediT authorship contribution statement

Conceptualisation: AH, DJL. Formal analysis: EV. Investigation: all authors. Data curation: EV. Visualization: EV. Writing – original draft: EV, AH. Writing – review & editing: all authors. Supervision – AH, DJL. Funding acquisition: AH, DJL.

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Data availability

Anonymised data are available only via application to the NHS Greater Glasgow & Clyde SafeHaven. Clinical samples are restricted for use under the ethical approvals obtained for their use.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Roche provided additional Cobas Liat influenza/SARS-CoV-2 point-of-care PCR testing platforms for this study. MEM has received research funding and support from Pfizer, Biomerieux, Shionogi, Infectopharm, and consulting fees from Tillots. AH has received research funding from UKRI, BSAC, Medical Research Foundation, Gates Foundation, and Pfizer unrelated to this work.

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Patient consent statement

Ethics approval was obtained from the North of Scotland Research Ethics Committee (Ref: 21/NS/1036) and the NHSGGC Biorepository (ref: 16/WS/0207) for storage of residual samples. Patients' written informed consent was obtained prior to the collection of biological samples.

Patient and public involvement and engagement statement

A patient and public involvement group reviewed the study protocol, participant information sheet and consent process. Their feedback informed refinements to participant-facing materials and helped ensure that recruitment and consent procedures were clear, proportionate and acceptable for acutely unwell hospitalised adults and consultees/proxies. Patients and public contributors were not involved in recruitment, data collection or analysis. The group included individuals with relevant lived experience of hospital care and were reimbursed for their time in line with NIHR guidance.

Data presentation

Results from this study were presented as posters at ESCMID Global, Vienna, Austria, 11–15th April 2025, and the 7th Respiratory Tract Infection Forum, Glasgow, UK, 3–5th February 2025, and comprised a chapter of Elen Vink's PhD thesis.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jinf.2026.106805.

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